

John Alvin Ambalong

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SUMMARY

Rails backend engineer focused on API design, service-layer architecture, and DevOps tooling. Currently shipping Pangu Terminal, a resource optimization platform deployed to DigitalOcean via Kamal 2, while completing an AAS in Software Design at Walla Walla Community College.

SKILLS

Backend: Ruby on Rails, Hotwire (Turbo / Stimulus), PostgreSQL, ActiveRecord, RESTful APIs, Service Objects, MVC, OOP

Frontend: JavaScript, HTML5, CSS3, Stimulus.js, Rails Views (ERB)

DevOps: Docker, Linux, DigitalOcean, Kamal 2, Git, PowerShell

Practices: Minitest, TDD, N+1 prevention, Service-layer architecture

PROJECTS

Pangu Terminal · Rails 8 · Hotwire · PostgreSQL · Docker · Kamal 2

panguterminal.ambalong.dev · github.com/jambalong/pangu-terminal

- Built a service-based planner that queries 10+ static cost tables across character and weapon upgrade paths, keeping game rules and business logic independently testable.
- Ships a versioned REST API where a single endpoint pair returns material deficits and farming sources, secured with SHA-256 hashed bearer tokens and per-key rate limiting.
- Inventory edits trigger Turbo Stream re-renders of the entire synthesis family, reflecting recalculated craftable counts without a page reload.
- Detects 3-to-1 synthesis opportunities from surplus lower-tier materials and reconciles EXP potion equivalence across tiers to verify plan completeness.
- Waveplate Optimizer estimates domain runs and stamina cost per deficit material by farming source, ranking source types by deficit coverage to surface the highest-value farm first.

EFI Secure Boot Compliance Script · PowerShell · ManageEngine

github.com/jambalong/efi-bootmgr-cert-check

- Deployed via ManageEngine to Windows 11 endpoints fleet-wide to audit whether each machine's boot manager binary is signed by the Windows UEFI CA 2023 certificate chain.
- Determined through research that Get-AuthenticodeSignature and Get-PfxCertificate return incorrect issuer data for EFI binaries; used X509Certificate.CreateFromSignedFile as the correct method after confirming a known PowerShell issue.

Ruby Chess Engine · Ruby

github.com/jambalong/chess

- Built a command-line chess engine in pure Ruby with no framework, modeling the full domain across Board, Piece, and Player classes with clean separation of responsibilities.
- Implemented legal move validation for all pieces including directional constraints and collision detection, with the board fully queryable at any point during a turn.
- Detected check and checkmate by deep-cloning the board state to simulate each candidate move, then verifying whether the king remains under attack after the move resolves.

EXPERIENCE

Information Technology Intern

Jan 2025 – Present

Walla Walla Community College

- Maintained 10+ computer labs and 100+ workstations supporting uninterrupted academic operations for students and faculty across campus.
- Analyzed lab usage data against term calendars, college closures, and scheduled events to produce summary reports informing leadership resource planning.
- Wiped 100+ decommissioned machines and managed the full surplus disposal process to meet data security compliance requirements before equipment left campus.
- Reimaged 60+ computers annually, restoring devices to deployment standard and returning equipment to full academic service within the same term.
- Audited 600+ technology assets across 65+ classrooms and campus facilities, establishing accurate departmental inventory records for the first time.

EDUCATION

AAS in Software Design

Expected June 2026

Walla Walla Community College